

Climate vulnerability in Southwark

Climate risk

Climate risk depends on three factors (Greater London Authority and Arup, 2024). The maps below apply them to heat in Southwark.

- hazard: weather and climate events that can cause harm, such as heatwaves; climate change alters how often, how long and how intensely they occur
- exposure: where people, property and other resources are, relative to a hazard
- vulnerability: how likely exposed people or property are to be harmed; reduced by the capacity of people and places to adapt

Urban heat island: hazard and exposure

Arup's UHeatRank, the GLA heat hazard map, ranks every London LSOA from 1 (coolest fifth) to 5 (hottest fifth) by modelled average summer air temperature, June to August 2018 to 2022 (Greater London Authority and Arup, 2024). Air temperature is closer to what people feel than land surface temperature.

Most of north and central Southwark is rank 4 or 5 (Figure 1a). Ranks fall towards the south; Dulwich and the large parks rank lowest. Dense buildings, hard surfaces and waste heat drive the effect: central London runs about 4.5°C hotter than its rural surroundings (Arup, 2023).

Greening is the most direct lever. Raising tree cover to 30% across 93 European cities could have prevented 2,644 deaths in summer 2015, about a third of those caused by urban heat islands (Iungman et al., 2023).

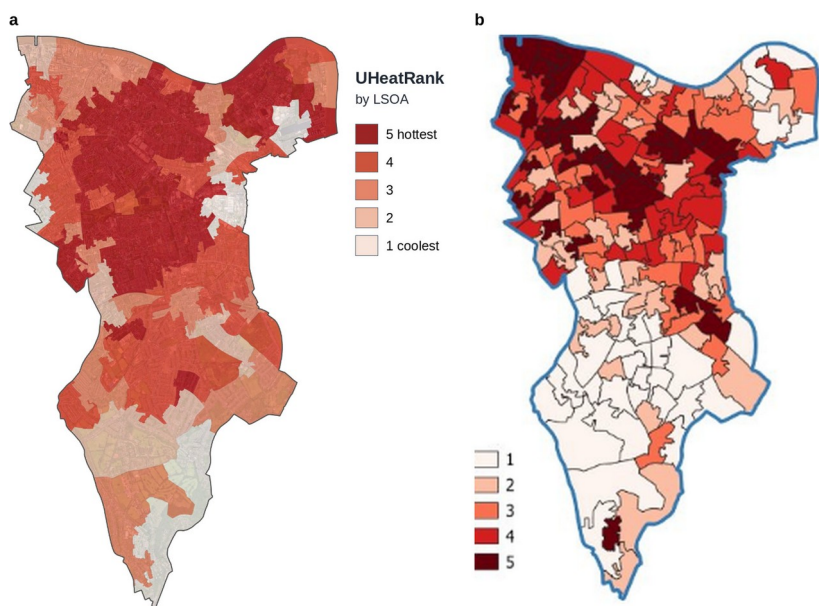


Figure 1. (a) UHeatRank by LSOA. Source: Greater London Authority and Arup (2024). (b) Neighbourhood heat vulnerability by LSOA, 1 (lower) to 5 (higher). Source: Sayers and Partners (2025).

Neighbourhood heat vulnerability

Vulnerability comes from how personal, environmental and social factors combine (Sayers and Partners, 2025):

- personal: age and health, which set sensitivity to heat
- environmental: green space, housing quality and building form, which raise or offset exposure
- social and institutional: income, inequality, social networks, neighbourhood cohesion and how institutions such as care homes operate, which shape the ability to adapt

Vulnerability is highest in the north-west and centre-north of the borough, with smaller clusters in the east and far south (Figure 1b). It does not simply follow heat: Rotherhithe ranks hot but scores low for vulnerability. The highest scores mark where hard urban form, homes poorly suited to heat (flats and top floors with limited ventilation) and residents less able to cope (older people, those with health conditions, low-income households) coincide. Many of these areas are among the most deprived, so heat risk and disadvantage reinforce each other.

Southwark's Hot Weather and Health JSNA confirms the pattern (Southwark Council, 2025):

- over half of Southwark is at risk of regular overheating at 1.5°C of warming

- up to 123,000 residents are at higher risk from heat because of underlying health conditions
- residents of the most disadvantaged wards worry about keeping their home cool at close to three times the rate of those in the least disadvantaged
- rough sleepers, older adults living alone and people with mental health, respiratory or cardiovascular conditions are over-represented among those most affected

Indoor monitoring bears this out. The Bureau of Investigative Journalism and the Urban Big Data Centre fitted sensors in 40 Southwark homes in summer 2023. Every home reached at least 25°C, the WHO maximum recommended indoor temperature for London, and homes on communal heating ran about 1°C hotter (Bureau of Investigative Journalism, 2023). In the early September heat, 85% exceeded 27°C (University of Glasgow, 2023).

Green cover

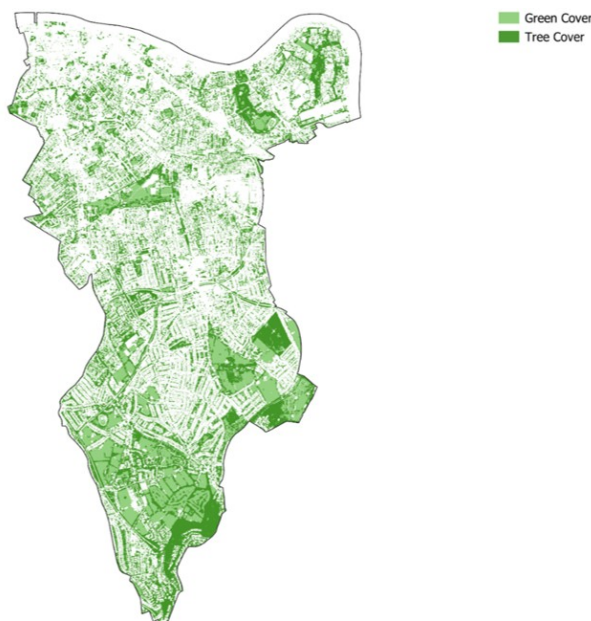


Figure 2. Green and tree cover, Southwark. Source: Southwark Council.

Green and tree cover is dense in the south and sparse across the north, apart from Burgess Park and Southwark Park. Set against Figure 1, the greenest areas are the coolest, and the least green are both hottest and most vulnerable. Placing canopy, planting and permeable surfaces where heat and vulnerability overlap directs cooling to the residents most exposed.

Estates

We have modelled heat vulnerability across Southwark's housing estates using the GLA method (Greater London Authority and Arup, 2024). It shows where building form and resident vulnerability combine to give the highest risk: a ranked basis for passive cooling, shading, ventilation and retrofit.

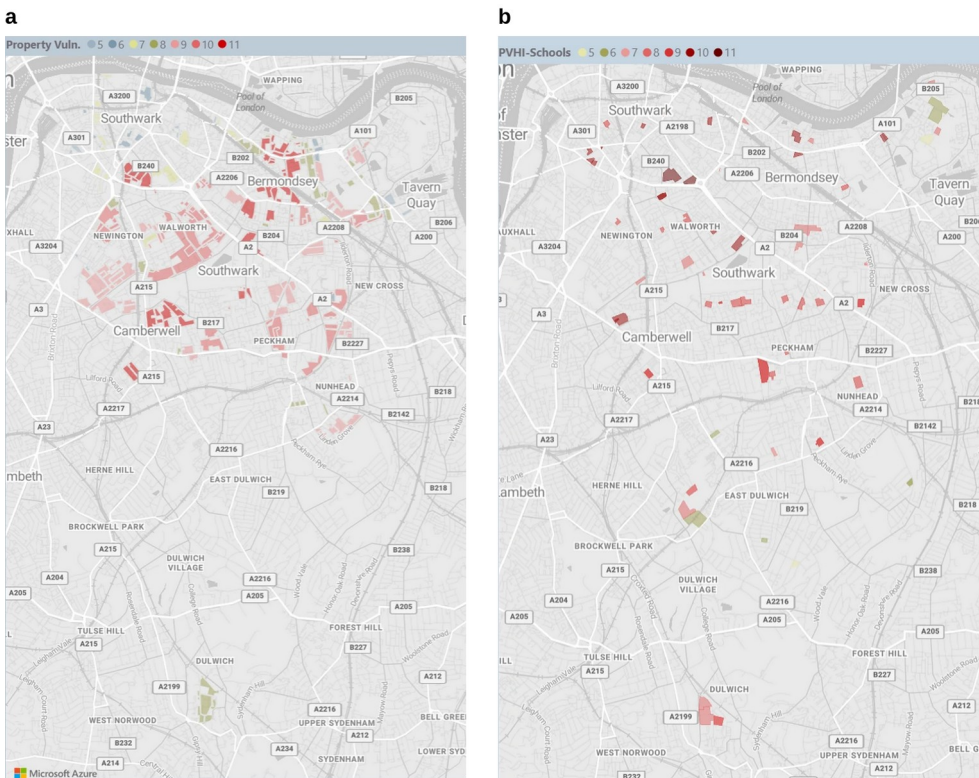


Figure 3. (a) Property heat vulnerability, council estates. Source: Southwark Council, GLA method. (b) School heat vulnerability index by site. Source: Southwark Council (2026). Scores run from 5 (lower) to 11 (higher).

The highest-scoring estates (Figure 3a) cluster in Walworth, Bermondsey, Camberwell and Peckham. The model informs Awaab's Law Phase 2, by showing where heat defects are likely to coincide with vulnerable residents, and the Decent Homes Standard, by identifying the homes most in need of retrofit and shading.

Council homes

Southwark is the largest council landlord in London. Much of its stock, including many purpose-built and high-rise flats, predates current ventilation and shading standards: the combination the

national assessment rates highest risk (Southwark Council, 2025). The Hot Homes pilot has shortlisted estates using the GLA and Arup data with social heat vulnerability data, and is due for implementation in spring or summer 2027.

National modelling for the Climate Change Committee compared adaptation packages rolled out to the 30% of neighbourhoods at highest risk. BCR is the benefit to cost ratio (Arup, 2026).

Package	Measures	BCR central	BCR high warming
A Foundational	shading, ventilation, insulation, blinds	0.1	below 1
B, C Enhanced	A plus green, blue and urban cooling	0.02 to 0.03	below 1
D Active cooling	air conditioning	3.2	6.3
E Integrated	passive plus targeted active cooling	3.0	6.0

Extending package E to the 45% of neighbourhoods at highest risk keeps heat deaths in 2059 just below today’s level, with a benefit to cost ratio of 2.6 (Arup, 2026).

Passive measures remain the right first step for the pilot: lower cost, faster to deliver, and free of the energy and emissions trade-offs of active cooling. Alone, they are unlikely to close the gap the borough’s heat data imply. The case for targeted active cooling in the highest-risk homes, once the pilot’s evidence is in, is likely to strengthen.

Schools

A schools heat risk assessment, carried out during the red heat-health alert of June 2026, ranks all 116 open schools using the GLA and Arup school heat vulnerability index, applied to each school's LSOA (Southwark Council, 2026; Greater London Authority and Arup, 2024).

- 61 schools are in the higher vulnerability tier, 37 moderate and 18 lower
- 75 sit in areas of elevated urban heat
- 83 have early years pupils, including 45 in the higher tier
- 29 have tree canopy below the borough lower quartile, including 17 in the higher tier

The darkest school sites (Figure 3b) combine elevated heat, many young or vulnerable pupils and little shade. They are the natural starting point for shading, tree planting and cooling ventilation within schools capital programmes, subject to funding, stock condition and school support. All schools will face excess heat in heatwaves.

National evidence points the same way. Days with classrooms at 35°C or more, when learning becomes very difficult, are projected to rise from 1.7 a year now to 2.9 at 2°C of warming and 7.8 at 4°C. Learning days lost to generally warmer temperatures rise from 6.7 to 8.2 and 11.4 (Department for Education, 2025). Under high warming by the 2080s, lost learning could reach 14 days a year, with children in southern England facing severe loss of cognitive performance for over 80% of the spring and summer months (Climate Change Committee, 2026).

The LLFA has also completed SuDS opportunity modelling for schools to reduce surface water flood risk.

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